A Numerical and Analytical Approach to Radioactive Decay

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We report on the solutions to coupled dual radioactive decay, both analytically and numerically. The numerical solution we implement is called Euler's method, which is a first-order method. We find that the shape of the curve is dependent on how the time constants of the nuclei compare to each other. Additionally, we explore the limitations and stability criteria of Euler's method on this system.

I. INTRODUCTION

Radioactive decay is a process in which an unstable nucleus emits by radiating particles [1]. There are many types of unstable nuclei, which makes it important to be able to predict how those nuclei would decay over time.

The example in this paper will take the case of dual decay. In dual decay, a nucleus of one type undergoes radioactive decay and turns into nucleus of another type. This other nucleus also decays, but not necessarily at the same speed. In the case of dual decay, the number of nuclei of both types are coupled. This makes modeling the system much more complicated.

II. THEORY AND EQUATIONS

This paper will explore Euler's method on the scenario described the differential equations shown in Eq. (1) and Eq. (2), where type A nuclei decay into type B nuclei, which also decay.

$$\frac{dN_A}{dt} = -\frac{N_A}{\tau_A}, \tag{1}$$
$$\frac{dN_B}{dt} = \frac{N_A}{\tau_A} - \frac{N_B}{\tau_B}. \tag{2}$$

$N_A(t)$ and $N_B(t)$ represent the number of nuclei of Type A and B, respectively while τ_A and τ_B represent a time constant for the decay for each type. Each decay constant tau affects the rate at which the nuclei decay such that a bigger decay constant requires a proportional larger time to result in the same amount of decay.

An attempt to normalize the equations were made such that $\overline{\tau_A} = \frac{\tau_A}{t0}$ and $\overline{\tau_B} = \frac{\tau_B}{t0}$. Setting $t0 = 1$ ultimately creates a very similar set of equations as the original one, except all the values a dimensionless. Any value utilized later in the paper is understood to be normalized.

To simplify the analysis, initial conditions were chosen for each type of nuclei. All of the latter equations and figures will be using initial conditions of $N_A(t=0) = 200$ and $N_B(t) = 5$

III. METHODOLOGY

The analytical solution can be found by treating the equations as a homogeneous linear time-invariant (LTI) system. which can be described as $\dot{X} = AX$, where X is a vector containing the required states and A is the state matrix. For this problem, the states will be N_A and N_B. Therefore, the state equation can be written as

$$\begin{bmatrix} \dot{N_A} \\ \dot{N_B} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\tau_A} & 0 \\ \frac{1}{\tau_A} & -\frac{1}{\tau_B} \end{bmatrix} \times \begin{bmatrix} N_A \\ N_B \end{bmatrix}$$

The solutions to this derivation can be found using the matrix exponential, which acts as the state transition. The solutions will take the form of $X = e^{At} * X(0)$ where $X(0)$ contains the initial values of the states. The matrix exponential can be solved for using the Cayley-Hamilton theorem, shown in Eq. (3) below

$$e^{At} = \alpha_0 I + \alpha_1 A, \tag{3}$$

where I is the identity matrix. The values for α_0 and α_1 can be found by solving the following system of equations.

$$e^{\lambda_1 t} = \alpha_0 + \lambda_1 \alpha_1 \tag{4}$$
$$e^{\lambda_2 t} = \alpha_0 + \lambda_2 \alpha_1, \tag{5}$$

where λ_1 and λ_2 are the eigenvalues of A. Solving for the eigenvalues, plugging them into the system of equations, and solving the system results in the following values for α_0 and α_1.

$$\alpha_1 = \frac{e^{-t/\tau_A} - e^{-t/\tau_B}}{1/\tau_B - 1/\tau_A} \tag{6}$$
$$\alpha_0 = e^{-t/\tau_A} + \frac{1}{\tau_A}\alpha_1 \tag{7}$$

Plugging these values into Eq. (3) results in the following solutions, after applying the state transition matrix to the initial conditions.

$$N_A = 200 e^{-t/\tau_A}, \tag{8}$$
$$N_B = 5e^{\frac{-t}{\tau_A}} + \left(\frac{205}{\tau_A} - \frac{5}{\tau_B}\right)\left(\frac{e^{\frac{-t}{\tau_A}} - e^{\frac{-t}{\tau_B}}}{\frac{1}{\tau_B} - \frac{1}{\tau_A}}\right) \tag{9}$$

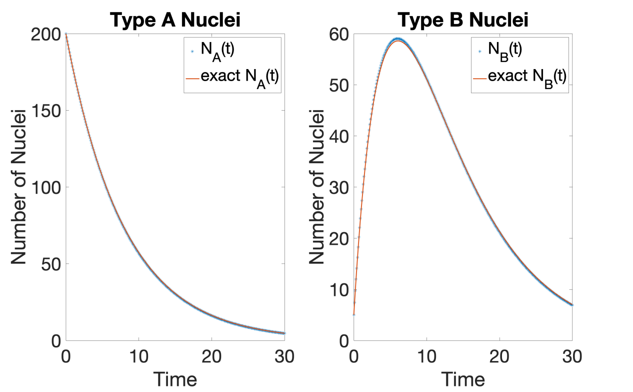

FIG. 1. Number of Type A and Type B Nuclei when $\tau_A = 8$, $\tau_B = 5$, Δt = 0.1

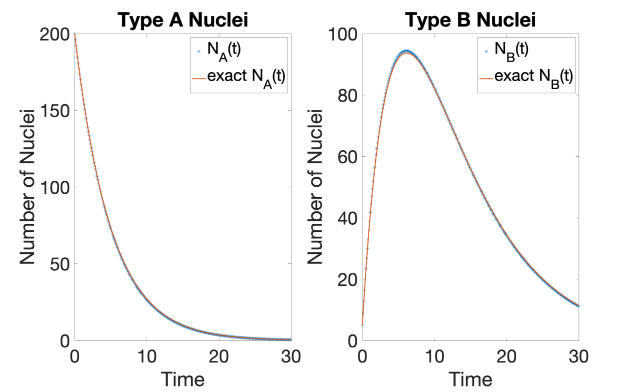

FIG. 2. Number of Type A and Type B Nuclei when $\tau_A = 5$, $\tau_B = 8$, Δt = 0.1

This solution; however, has a discontinuity when $\tau_A = \tau_B$. This can be remedied by setting $\tau_A = \tau_B = \tau$ and finding a new analytical solution. This creates a state matrix which has a repeated eigenvalue, which creates an under-determined system for Eq. 4 and Eq. 5. One can generate another equation by taking the time derivative of Eq. 4.

Below is the analytical solution found using this methodology for $\tau_A = \tau_B$.

$$N_A = 200e^{-t/\tau}, \tag{10}$$

$$N_B = 200(\tfrac{t}{\tau}e^{-t/\tau}) + 5e^{-t/\tau} \tag{11}$$

Euler's Method is a first-order method to approximate solutions to differential equations . It uses a first order Taylor series to approximate the next term of the function [1]. In general, it is written as

$$N(t + \Delta t) \approx N(t) + \frac{dN}{dt}\Delta t \tag{12}$$

This was implemented in MatLab using a for loop to iterate to each next time step. Results are plotted for different cases with the exact and estimated values sharing a plot.

IV. RESULTS

Figure 1 shows a case such that $\tau_A > \tau_B$. Figure 1 shows a case such that $\tau_A < \tau_B$. Figure 1 shows a case such that $\tau_A = \tau_B$. The difference between these cases are discussed in the next section.

Additionally, Figure 4 and Figure 5 show cases where τ_A is decreased to a point where it results in oscillatory and unstable solutions. More specifically, the plots show when $\tau_A = \frac{1}{2}\Delta t$ and $\tau_A < \frac{1}{2}\Delta t$, respectively.

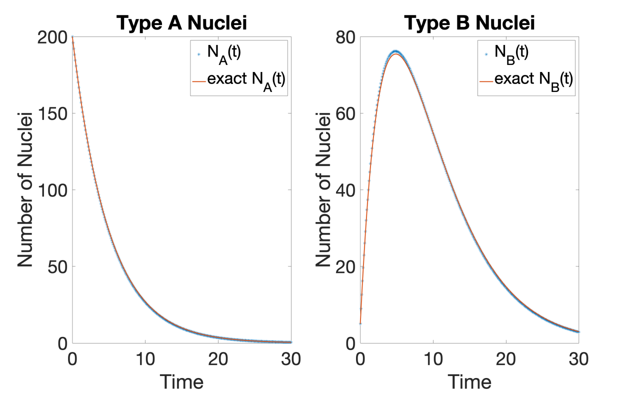

FIG. 3. Number of Type A and Type B Nuclei when $\tau_A = 5$, $\tau_B = 5$, Δt = 0.1

V. DISCUSSION

Comparisons between the numerical approximation and the analytical solutions can be made in Figures 1, 2, and 3. It seems that the approximation will overshoot the value when the concavity is negative and un-

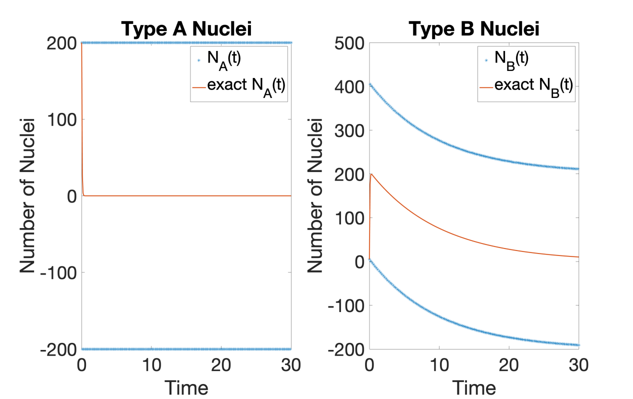

FIG. 4. Number of Type A and Type B Nuclei when $\tau_A = 0.05$, $\tau_B = 10$, Δt = 0.1

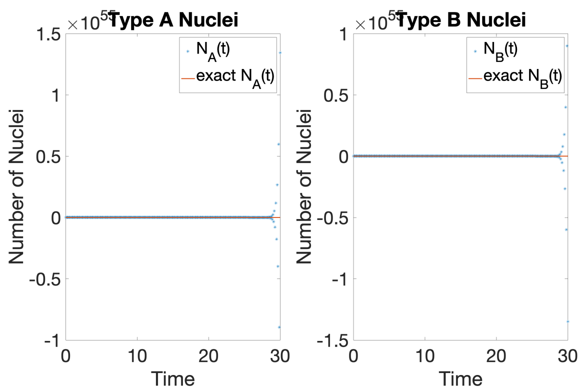

FIG. 5. Number of Type A and Type B Nuclei when $\tau_A = 0.04$, $\tau_B = 10$, Δt = 0.1

dershoot the value when the concavity is positive. This makes sense because concavity is a second-order term, and Euler's method eliminates terms higher than first order. This makes sense because eliminating a negative term will cause the approximation to be higher than the exact value, and vice versa for positive terms.

Differing the ratio between τ_A and τ_B causes B to peak at higher values. The greater τ_B is relative to τ_A, the greater the peak value is. This aligns with expectations, as increasing the time constants will slow down the decay. If Type B Nuclei decay slower, then it allows more of Type A Nuclei to turn into Type B Nuclei before the Type B decay dominates the Type A to Type B decay.

Differing the ratio between τ_A and Δt shows stability limitations with Euler's method. When $\tau_A = \frac{1}{2}\Delta t$, as depicted in 4, the Euler's method solution is perfectly oscillatory. The values oscillate between 200 and -200, without decaying or blowing up to large values. This is no longer true once τ_A decreases slightly in value, as depicted in 5, where values get many magnitudes of order larger. This can be seen to also affect the solution of the Type B Nuclei even though τ_B is sufficiently large. This means that the implementation of Euler's method needs to carefully regard all variables that could effect the stability of the solution.

VI. CONCLUSIONS

By analyzing the numerical and analytical solutions, we get some insight on the physics of radioactive decay, as well as the benefits and limitations of Euler's method. Namely, that characteristics of the curve are highly dependent on the time constants for each type of nuclei. In the real world, nuclei of different atoms will have different time constants; therefore, we can expect different decay behavior for each one. For dual decay, we can expect that the number of secondary nuclei will peak at different values, depending on these constants. Additionally, we see that Euler's method is a very simple numerical method that is very easy to implement, but it is subject to noticeable error and instability.

Further studies could implement changing the initial conditions to see how this effects the behavior of the curve. Another study could also study semi-implicit or implicit methods or higher order methods and compare the errors between the numerical and analytical solutions.

[1] N. Giordano and H. Nakanishi, *Computational Physics* (Pearson Prentice Hall, Upper Saddle River NJ, 2006).